Theory Q5
Air cushion (10 points),
Solution and Marking Scheme

A.1 **0.5pt**	From $c_P - c_V = r$ and $c_P/c_V = \gamma$: $$c_P = \frac{\gamma r}{\gamma - 1} = \frac{7}{2} r \approx 1005 \ \frac{\mathrm{J}}{\mathrm{kg \cdot K}}, \qquad c_V = \frac{r}{\gamma - 1} = \frac{5}{2} r \approx 718 \ \frac{\mathrm{J}}{\mathrm{kg \cdot K}}.$$	
	$c_V = r/(\gamma - 1) \approx 718$ J/(kg·K)	**0.25**
	$c_P = \gamma r/(\gamma - 1) \approx 1005$ J/(kg·K)	**0.25**

A.2 **0.5pt**	Equilibrium of the platform: $(P_0 - P_{atm})\frac{\pi D^2}{4} = W$, so $$\frac{P_0}{P_{atm}} = 1 + \frac{4W}{P_{atm}\pi D^2} \approx 7.18.$$	
	$P_0/P_{atm} = 1 + 4W/(P_{atm}\pi D^2)$	**0.25**
	$P_0/P_{atm} = 7.18 \pm 0.05$	**0.25**

B.1 **0.5pt**	Take the region swept by the front between t and $t + \Delta t$: its mass grows from $\rho_0 A c\,\Delta t$ to $\rho A c\,\Delta t$, and the net inflow is $\rho v A\,\Delta t$. Hence $(\rho - \rho_0)c = \rho v$, $$v = c\left(1 - \frac{\rho_0}{\rho}\right).$$	
	Mass conservation written correctly	**0.3**
	$v = c(1 - \rho_0/\rho)$	**0.2**

B.2 **1.0pt**	The momentum theorem for the gas set in motion: $(P - P_0)A = \frac{\mathrm{d}p}{\mathrm{d}t} = v\frac{\mathrm{d}m}{\mathrm{d}t} = \rho_0 A c v$. With B1, $$P - P_0 = \left(1 - \frac{\rho_0}{\rho}\right)\rho_0 c^2.$$	
	Momentum theorem used (also credited with ρ in place of ρ_0)	**0.5**
	$P - P_0 = (1 - \rho_0/\rho)\rho_0 c^2$	**0.5**

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B.3 **0.5pt**	$c^2 = \frac{\rho}{\rho_0}\frac{P-P_0}{\rho-\rho_0}$; for a continuous change $c^2 = \frac{\mathrm{d}P}{\mathrm{d}\rho}$. The flow is adiabatic, $P\rho^{-\gamma} = \text{const}$, so $\frac{\mathrm{d}P}{\mathrm{d}\rho} = \gamma\frac{P}{\rho} = \gamma r T$ and $$c = \sqrt{\gamma r T}.$$	
	Method 1: the correct answer $c = \sqrt{\gamma r T}$	**0.5**
	(alternative) Method 2: the continuous limit $\rho \to \rho_0$ used (0.2); the derivative taken at constant entropy (0.2); $c = \sqrt{\gamma r T}$ (0.1). With a wrong numerical factor in c, B3 and B4 earn 0 and later items are marked consistently.	***0.5***

B.4 **0.5pt**	$$c_0 = \sqrt{\gamma r T_0} \approx 347\ \mathrm{m/s}.$$	
	$c_0 = (347 \pm 3)$ m/s	**0.5**

B.5 **1.0pt**	With $\rho = P/(rT)$: $\frac{\mathrm{d}m}{\mathrm{d}t} = \frac{PAcM}{rT} = \frac{\gamma AMP}{c_0}\sqrt{\frac{T_0}{T}}$. Adiabatically $T^\gamma P^{1-\gamma} = \text{const}$, $P = P_0(T/T_0)^{\gamma/(\gamma-1)}$, hence $$\frac{\mathrm{d}m}{\mathrm{d}t} = \frac{P_0 A \gamma}{c_0} M \left(\frac{T}{T_0}\right)^{\frac{\gamma+1}{2(\gamma-1)}}.$$	
	$\mathrm{d}m/\mathrm{d}t = PAcM/(rT)$	**0.1**
	$\mathrm{d}m/\mathrm{d}t = \frac{\gamma AMP}{c_0}\sqrt{T_0/T}$	**0.2**
	$P = P_0(T/T_0)^{\gamma/(\gamma-1)}$	**0.2**
	Final expression	**0.5**

B.6 **0.5pt**	$$v = Mc = M\sqrt{\gamma r T}.$$	
	$v = M\sqrt{\gamma r T}$	**0.5**

B.7 **1.5pt**	With v and c_P: $rT\left(\frac{\gamma M^2}{2} + \frac{\gamma}{\gamma-1}\right) = \text{const}$, i.e. $T((\gamma-1)M^2+2) = \text{const} = 2T_0$, so $$\frac{T}{T_0} = \frac{2}{2+(\gamma-1)M^2}.$$	
	$T((\gamma-1)M^2+2) = \text{const}$	**0.5**
	$T((\gamma-1)M^2+2) = 2T_0$	**0.8**
	$T/T_0 = 2/(2+(\gamma-1)M^2)$	**0.2**

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C.1 **1.0pt**	Combining B5 and B7, $$\frac{\mathrm{d}m}{\mathrm{d}t} = \frac{P_0 A\gamma}{c_0} M \left(1 + \frac{\gamma-1}{2} M^2\right)^{-\frac{\gamma+1}{2(\gamma-1)}},$$ which increases on $[0;1]$. Setting the derivative with respect to M to zero: $1 + \frac{\gamma-1}{2}M^2 - \frac{\gamma+1}{2}M^2 = 0$, i.e. $$M = 1.$$	
	Derivative $\partial(\mathrm{d}m/\mathrm{d}t)/\partial M$ computed correctly (for one's own expression)	**0.5**
	$M = 1$	**0.5**

C.2 **1.0pt**	For $M = 1$: $T = \frac{2T_0}{1+\gamma} = 250$ K; by Poisson's law $P = P_0 \left(\frac{2}{1+\gamma}\right)^{\gamma/(\gamma-1)} = 3.83 \cdot 10^5$ Pa and $\rho = \frac{P_0}{rT_0}\left(\frac{2}{1+\gamma}\right)^{1/(\gamma-1)} = 5.34\ \mathrm{kg/m^3}$; $v = \sqrt{\frac{2\gamma r T_0}{\gamma+1}} = 317$ m/s. With the delivered flow I, $\frac{\mathrm{d}m}{\mathrm{d}t} = I\frac{P_{atm}}{rT_0} = \frac{\gamma P_0 A}{c_0}\left(\frac{2}{1+\gamma}\right)^{\frac{\gamma+1}{2(\gamma-1)}}$ and $A = \pi D\varepsilon$: $$\varepsilon = \frac{I}{c_0 \pi D}\frac{P_{atm}}{P_0}\left(\frac{1+\gamma}{2}\right)^{\frac{\gamma+1}{2(\gamma-1)}} = 0.046\ \mathrm{mm}.$$ (These values are recomputed from the formulas with $P_0 = 7.18 P_{atm}$; taking $P_0 = P_{atm}$ instead gives $P = 0.53 \cdot 10^5$ Pa, $\rho = 0.74\ \mathrm{kg/m^3}$, $\varepsilon = 0.33$ mm, which the scheme also accepts.)	
	Formulas for T, P, ρ, v, ε (0.1 each)	**0.5**
	Values with $P_0 = 7.18 P_{atm}$: $T = 250$ K, $P = 3.83 \cdot 10^5$ Pa, $\rho = 5.34\ \mathrm{kg/m^3}$, $v = 317$ m/s, $\varepsilon = 0.046$ mm (0.1 each)	**0.5**
	(alternative) Or, with $P_0 = P_{atm}$: $T = 250$ K, $P = 0.53 \cdot 10^5$ Pa, $\rho = 0.74\ \mathrm{kg/m^3}$, $v = 317$ m/s, $\varepsilon = 0.33$ mm (0.1 each). (The published scheme lists ε of the two cases the other way round.)	***0.5***

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D.1 **1.5pt**	As in C2, but first find M from $$\frac{\mathrm{d}m}{\mathrm{d}t} = \frac{P_0 \pi \varepsilon D \gamma}{c_0} M \left(1 + \frac{(\gamma-1)M^2}{2}\right)^{\frac{\gamma+1}{2(1-\gamma)}},$$ numerically or with $M \ll 1$: $M = 0.055$, then $T = 299.8$ K, $P = 7.24 \cdot 10^5$ Pa, $\rho = 8.41\ \mathrm{kg/m^3}$, $v = 18.9$ m/s. (Recomputed from the formulas with $P_0 = 7.18 P_{atm}$; the published scheme lists $T = 297$ K, $P = 6.98 \cdot 10^5$ Pa, $\rho = 8.20\ \mathrm{kg/m^3}$, $v = 18.8$ m/s and accepts its tolerance band. With $P_0 = P_{atm}$: $M = 0.44$, $T = 289$ K, $P = 0.89 \cdot 10^5$ Pa, $\rho = 1.07\ \mathrm{kg/m^3}$, $v = 149$ m/s.)	
	Method 1: the equation for M solved numerically	**0.5**
	Values (0.2 each): $M = 0.055 \pm 0.005$, $T \approx 297$–300 K, $P \approx (6.98\text{–}7.24) \cdot 10^5$ Pa, $\rho \approx 8.2$–$8.4\ \mathrm{kg/m^3}$, $v = (18.8 \pm 1.0)$ m/s	**1.0**
	(alternative) Method 2: the justified approximation $M \ll 1$ (0.5) with the values for $P_0 = P_{atm}$ (0.2 each): $M \approx 0.44$, $T \approx 276$–289 K, $P \approx (0.75\text{–}0.89) \cdot 10^5$ Pa, $\rho \approx 0.95$–$1.07\ \mathrm{kg/m^3}$, $v = (147 \pm 10)$ m/s	***1.5***

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